VU THANH DUC

AI Engineer | Python · PaddleOCR · LLMs · Whisper · .NET · Node.js · AWS · Docker
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SUMMARY

AI and back-end engineer who builds and ships end to end: a GPU-accelerated OCR, LLM translation and speech pipeline in Python, a microservices speech-analysis platform I led, and .NET production systems at FPT Software where I fixed race conditions and cut lookup latency with Redis. Hands-on with JWT/OAuth2 authentication and with deploying and configuring my own servers on DigitalOcean and AWS. Seeking an AI Engineer role, with growing focus on security and cloud reliability.

TECHNICAL SKILLS

Languages	Python, C#, TypeScript, JavaScript (ES2023+)
AI / ML	PaddleOCR, OpenCV, Faster Whisper, Pyannote.audio, Sentence Transformers, spaCy, OpenAI API, Google Gemini, Mistral/Cohere LLM APIs, Ollama (local LLMs)
Security & Auth	JWT, OAuth2, retry and rate-limit handling, transactions and locking for data integrity
Cloud & DevOps	DigitalOcean, AWS (S3, SQS, RDS), Azure, Docker, Jenkins CI/CD, SonarQube
Backend	Python, .NET Core, Node.js (Express), RESTful APIs, Microservices
Databases	SQL Server, MySQL, PostgreSQL, MongoDB, Redis
Frontend	React.js, Next.js, Vue.js, Redux Toolkit, TailwindCSS
Tools	Git/GitHub, Jira, xUnit/Moq, Agile/Scrum

PROJECTS

Subtitle OCR, Translation & TTS Pipeline — AI automation
Jun 2026 – Present

Python · OpenCV · PaddleOCR (GPU) · Mistral / Cohere LLMs · Ollama · TTS
Personal project, used by friends; not publicly released

- Built a two-stage pipeline: video to Chinese subtitles (OCR of burned-in text), then Vietnamese SRT plus a timeline-synced TTS mp3. Runs on a laptop GTX 1660 Ti at ~3.1x real time (≈19 min for a 1-hour video, OCR stage).
- OCR: two-pass design (detect subtitle segments, then one OCR call per segment), consensus voting across frames to avoid fabricated lines, 10 fps recovery pass for missed cues, checkpoint/resume for long jobs.
- Translation: Google Translate first pass, cloud LLMs refine only cues a scoring step marks weak; domain glossary, quality gate, retry/backoff on rate limits, daily token budget, and automatic fallback to local models (Ollama Qwen3, opus-mt).
- Results: 94.3% clean cues on a 14,996-cue file; ~23,700 cues translated and ~28,000 TTS segments produced to date.

OratorAI — AI Presentation Analysis Platform
Jan – Apr 2026

Team Leader & Back-end Developer · Team of 5 · Academic project

- Designed a microservices system for presentation analysis (Faster Whisper, Pyannote.audio, Sentence Transformers + spaCy); used Amazon SQS to decouple AI workers from the API for horizontal scaling.
- Integrated Google Gemini to auto-generate structured performance reports.

Also (Team Leader, team of 5): AImpact — AI personal finance app (Grok, PayOS) · Quizlingo — adaptive English quiz app (OpenAI) · EVCare — EV service management system (Gemini, PayOS)

EXPERIENCE

Back-end & Front-end Developer — FPT Software
Jan 2025 – Jun 2026

Smart Inventory & Order Tracking System (APAC retail client)
May 2025 – Oct 2025

.NET Core · SQL Server · Redis · Docker

- Built APIs for order placement and real-time inventory sync; resolved low-stock race conditions with transactions and locking on SQL Server (no overselling in flash-sale trials).
- Added Redis caching, cutting stock-lookup response time from ~800ms to under 150ms; wrote xUnit/Moq tests covering 75%+ of key business logic.

Citizen Management Platform (Japan)
Nov 2025 – Jun 2026

C# (.NET Core) · Vue.js · SQL Server · Jenkins CI/CD

- Built and maintained C# (.NET) services for a government citizen management platform deployed in Japan; developed a shared Vue.js component library adopted across the product.

Internal Resource & Sprint Tracking Tool
Jan 2025 – Apr 2025

Vue.js · Node.js (Express) · MongoDB · Docker

- Built Express REST APIs and a Vue.js dashboard syncing Jira tasks and SonarQube coverage via webhooks; cut weekly report time by ~60%; used by 100+ people in the unit.

Freelance Full-Stack Developer — Solo projects
Jan 2026 – Sep 2026

- **Happy Furniture** (happyfurniturevn.com): React storefront, Next.js admin, .NET REST API; shipped live from requirements to deployment, 30+ products.
- **Restaurant table-side ordering system:** QR ordering with real-time routing to a kitchen display (React, Node.js, MySQL, WebSocket), admin panel and daily sales reports.
- **I8 Studio** (i8.com.vn): live showcase site for a TVC production studio (Next.js, Prisma, Tailwind CSS v4).

EDUCATION

FPT University — Bachelor of Software Engineering *Sep 2021 – May 2026*
English: Working proficiency — comfortable reading technical documentation.